

Oogiri Task: Preventing Daytime Drowsiness by Generating Humorous Captions for Images

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Abstract

Sustaining high arousal levels during daytime tasks, such as office or academic work, remains a persistent challenge. This study explores a method that does not depend on specific environmental conditions and invokes amusement to enhance physiological arousal. As such, we introduce Oogiri — a form of improvisational comedy rooted in Japanese media culture, in which participants spontaneously respond to given prompts. To evaluate its effectiveness, we conducted controlled studies comparing subjective drowsiness before and after Oogiri and control task engagement, using Vigilance tasks. The results tentatively suggest a possible role for Oogiri in mitigating arousal decline, although further validation is needed.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Computers in other domains**; **Law, social and behavioral sciences**.

Keywords

Sleepiness, Vigilance Task, Emotion, Amusement

ACM Reference Format:

Ryo Yoshizawa, Koki Honda, and Rui Fukui. 2025. Oogiri Task: Preventing Daytime Drowsiness by Generating Humorous Captions for Images. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754390>

1 Introduction

Drowsiness during work or study is a common issue with serious implications in fields such as transportation, clinical care, and manufacturing [9, 14, 19]. While insufficient sleep is a primary cause [21], a predictable drop in arousal — known as the "afternoon dip" — also occurs due to circadian rhythms, typically in mid-afternoon [5, 8]. Since this period coincides with core working and study hours, maintaining high arousal levels and preventing daytime dozing is essential for sustaining task performance.

In automotive and clinical domains, studies have examined the use of auditory [23], olfactory [15, 18], electrical [24] stimuli or light [7] to help drivers remain alert.

However, existing arousal-enhancement techniques pose challenges when applied to shared environments such as offices. Stimuli involving sound or smell can improve arousal levels, but may disturb people nearby. While fixed-installation methods are suitable for controlled environments such as vehicle cabins or individual workstations, they are impractical in shared spaces like meeting rooms, where users cannot freely change the environment. In addition, scent, sound, and electrical stimuli may cause discomfort to the user. Other strategies, such as exercise [4, 13] and chewing [11], are often restricted by social norms and etiquette in offices and schools. Accordingly, there is a clear need to induce a high level of arousal and prevent dozing off in ways that are unobtrusive, socially acceptable, and feasible for use anytime and anywhere.

Given these constraints, this study outlines three essential criteria for effective arousal-enhancement interventions. First, it should sustain its drowsiness-prevention effect over extended periods, aligning with the goal of supporting productivity in work and academic settings. Second, the intervention must function independently of environmental constraints or bodily attachments, ensuring availability in a wide range of everyday situations. Third, to support long-term use, the intervention must avoid discomfort and instead deliver stimulation that users find pleasant.

This study aims to develop and validate a method to prevent drowsiness according to specified requirements. Specifically, we propose a technique to reduce drowsiness and assess its effectiveness through experimentation.

2 Proposed Approach

This section introduces the conceptual basis and proposed method for preventing drowsiness, aiming to support the overall objective of enhancing arousal during daily activities.

2.1 Arousal-Valence Model

This study argues that increasing arousal alone is insufficient to sustain long-term use; inducing pleasant emotions is also necessary. We employ a model that combines arousal and valence to characterize users' emotional state, identify a target affective condition, and design strategies to facilitate transitions toward this state. Russell's circumplex model of affect, a well-established multidimensional framework in affective psychology and human-computer interaction research, positions valence on the horizontal axis and arousal on the vertical axis [16]. Our objective is to simultaneously elevate arousal and induce positive affect by guiding the user's emotional state into the model's first quadrant. This quadrant includes emotions such as surprise, excitement, happiness, joy, and amusement [16, 17], with a particular focus on amusement as the target emotional state for this study.



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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754390>

2.2 A Humor-Based Task: Oogiri

We introduce a method for preventing drowsiness based on affective stimulation using Oogiri, a Japanese humor-based prompt-response activity like Figure 1. Oogiri is popular across amateur and professional communities and is accessible through various online platforms. Similar humor-based games also exist in English-speaking cultures (e.g., Caption This [6]). We hypothesize that generating humorous responses during Oogiri evokes amusement — an affective state characterized by high arousal and positive valence — and thereby helps prevent drowsiness. By implementing Oogiri as a mobile application, users can engage with the task on their personal devices, aligning with the operational constraints of our study environment. This research evaluates the effectiveness of Oogiri tasks in mitigating drowsiness.



Figure 1: Example of Oogiri Task

This study makes two key contributions as follows.

- Provides an empirical evaluation of the Oogiri task designed to increase arousal and reduce drowsiness.
- Presents the design of an experimental environment capable of inducing a low arousal state, which is necessary for evaluating interventions aimed at counteracting drowsiness.

3 Methodology

To assess whether Oogiri can serve as an effective countermeasure against daytime drowsiness, we conducted a two-phase experimental study involving human participants.

The preliminary experiment aimed to establish a procedure for reliably inducing drowsiness and to validate a method for evaluating the arousal-sustaining effects of a task. In the main experiment, we modified the preliminary experimental procedure using findings from the preliminary study and tested whether performing Oogiri tasks keeps arousal levels high. This section details the experimental environment, apparatus, and procedures shared across both experiments. All protocols were reviewed and approved by the Ethics Review Committee of the University of Tokyo (Approval No. 24-439).

3.1 Experimental environment

This subsection describes the controlled experimental environment designed to induce drowsiness and evaluate variations in arousal levels. The setup aimed to reduce sensory stimulation and promote

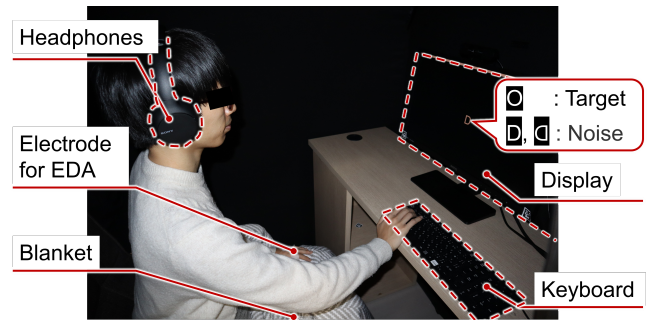


Figure 2: Experimental environment

physical comfort. Participants were seated in a reclining chair inside a black tent, which minimized visual input. Auditory stimuli were masked by playing white noise through headphones. To reduce physical tension and facilitate relaxation, participants removed their shoes and wore slippers. A blanket was placed over their laps to prevent discomfort caused by cold sensations in the lower limbs and to maintain a stable thermal environment. The full experimental setting is illustrated in Figure 2.

3.2 Vigilance Task

In this study, a vigilance task was implemented with two objectives: to elicit a reduction in arousal and to assess arousal level through behavioral performance, based on experimental paradigms used in previous studies on alleviating daytime sleepiness [10]. The task was adapted from the protocol developed by Temple et al. [20], which effectively induces lowered arousal within a short duration. During the task, participants observed a continuous sequence of letters (O, D, and Q: mirrored D in Figure 2) presented against a black background. All letters appeared in white. The target stimulus — the letter 'O' — was presented with a probability of approximately 10%. Participants responded to the target by pressing a key ('q') in the preliminary experiment or clicking a mouse in the main experiment. A computer system automatically recorded both stimulus onset times and response timestamps.

3.3 Physiological and Subjective Measures

This study assessed changes in arousal through subjective ratings and vigilance task performance. Additionally, skin conductance and electrocardiogram (ECG) were recorded as physiological indicators generally used to evaluate arousal in prior research on drowsiness assessment [1, 2]. Both measures were obtained using the biosignalsplux system and OpenSignals (r)evolution software (PLUX Wireless Biosignals S.A., Lisbon, Portugal).

Subjective measures included a participant's arousal and valence ratings. The arousal scale was a Japanese translation of the Karolinska Sleepiness Scale, a widely used subjective arousal [12], developed by Nakayama et al. [22]. The specific measurement procedures are detailed within each experiment's description.

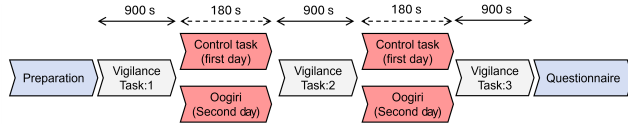


Figure 3: Preliminary experiment flow

3.4 Procedure

Participants completed the vigilance task for a predefined number of trials in each experiment. Between vigilance sessions, they performed intermediate tasks consisting of an Oogiri task and a control task. For both tasks, participants viewed images on the display and entered text responses into designated fields according to the experimenter's instructions. The control task was designed to resemble the Oogiri task in structure but excluded elements intended to evoke amusement, allowing for a comparison of arousal-sustaining effects. A Python-based system presented task screens on the display to administer a vigilance task and an intermediate task. Participants provided responses via mouse or keyboard.

4 Preliminary Experiment

This preliminary experiment examined the conditions needed to induce participant drowsiness and tested task presentation methods to evaluate the effect of the Oogiri task. The experiment included 3 graduate students (2 males, 1 female; mean age 24, SD 0.8; mixed gender).

4.1 Experimental Procedure

The vigilance task was performed 3 times, each lasting 900 s. Between sessions, participants completed intermediate tasks (Figure 3), which included an Oogiri task and a control task. In the Oogiri task, participants viewed a single image and composed a humorous caption. In the control task, participants described the image to enable visualization by someone unfamiliar with it (image description task). Participants recorded their responses via keyboard in both tasks. To minimize fluctuations in arousal caused by circadian factors and experimental fatigue, each intermediate task was administered on separate days at consistent times. All participants completed the control task on the first day and the Oogiri task on the second day. This sequence was selected to prevent the influence of creative thinking induced by the Oogiri task from affecting performance in the control task. Upon finishing each experimental session, participants indicated temporal changes in their subjective arousal by graphing on a sheet displaying the nine-point Karolinska Sleepiness Scale using a pen.

4.2 Experimental Results

Physiological data from Participant 1 in the image control task condition were unavailable due to a communication failure between the biosignalsplux device and the PC. Firstly, we report the results of the vigilance task.

Vigilance task performance is commonly evaluated based on metrics related to lapses and reaction times [3]. To assess pronounced drowsiness, this study employed metrics related to lapses. We quantified performance using the proportion of hits during 60 s, defined

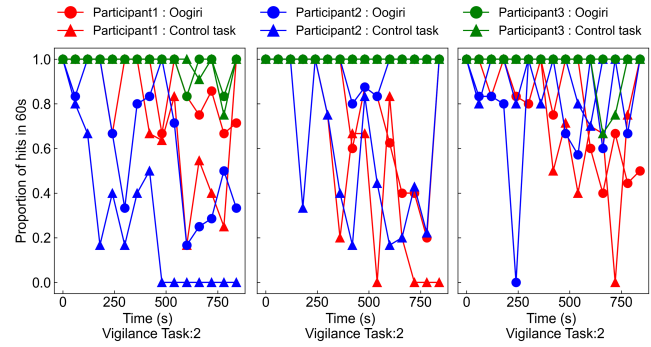


Figure 4: Time-Series Vigilance Task proportion of hits in each participant and condition

as the percentage of target stimuli ('O') presented at 60 s intervals to which participants responded within 1 s. Figure 4 presents the temporal changes in the proportion of hits during each vigilance task session, across all conditions and participants. Participants 1 and 2 started missing targets within 300 s of starting the task. The proportion of hits decreased over time, except near the task's end. These trends were observed across all sessions of the vigilance task. Participant 3 maintained target detection for the first 600 s but exhibited minor misses subsequently. Subjective arousal measures, described later, corroborate this pattern, indicating declines below the "Sleepy" threshold during periods of reduced detection. These findings confirm that the vigilance task, as implemented, reliably induces sleepiness under the experimental conditions.

In the Oogiri task condition, Participant 2 exhibited a temporal association between subjective arousal and performance on the vigilance task: periods of decreased self-reported arousal corresponded with a lower proportion of hits (Figure 5).

Nevertheless, physiological indicators did not consistently track fluctuations in arousal state. During the second half of the first vigilance task, the proportion of hits dropped to near zero due to repeated omissions, suggesting severely reduced arousal. In contrast, skin conductance increased during this period. Likewise, the heart rate remained stable during the second vigilance task, even as subjective arousal declined. These dissociations indicate that standard physiological indicators may not reliably index moment-to-moment changes in arousal state under these task conditions. In addition, during the third vigilance task, both subjective arousal and the proportion of hits remained stable at relatively high levels, whereas skin conductance showed a lower level.

In contrast, the proportion of hits in the vigilance task closely tracked fluctuations in subjective arousal, supporting their use as a reliable proxy for assessing the effectiveness of task-based interventions in preventing arousal decline. Consequently, we use the vigilance task performance as the primary measure of arousal in the main experiment based on these findings.

Two methodological limitations were found in this setup and procedure for assessing the impact of the Oogiri task. First, presenting only a single image during the task may have introduced variability in participant engagement. Some participants might have struggled to generate responses, while others may have completed their

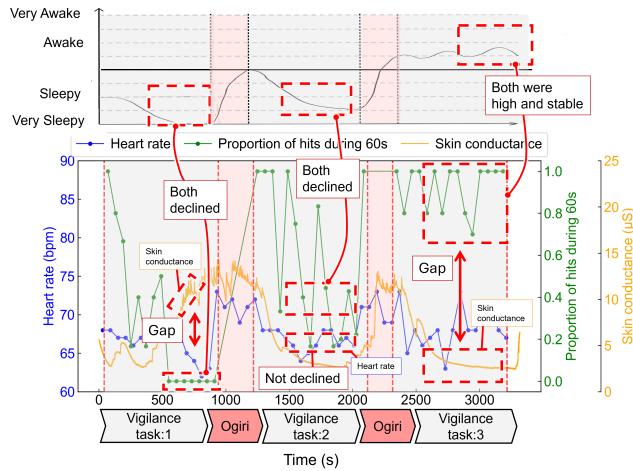


Figure 5: Participant 2: Time-Series of Subjective Ratings (Top), Skin Conductance, Heart Rate, and proportion of hits in 60s in Oogiri Condition

answers quickly and remained inactive for the rest of the session. These situations could have led to discomfort or unintended periods of rest, potentially reducing the task’s effectiveness in eliciting amusement. Second, the control task used as a control task may not have been fully neutral. The process of considering how to describe an image clearly in words might offer a degree of enjoyment in itself. This raises concerns about its suitability as a baseline when evaluating amusement-related effects.

5 Main Experiment

5.1 Experimental Procedure

The vigilance task is identical to those employed in the preliminary study. Several elements of the experimental procedure were modified based on findings from that preliminary experiment. For both Oogiri and control task, a set of more than ten images was prepared in advance. Participants could skip an image by pressing a button if they felt unable to generate a response. Upon submitting a response, a new image was automatically displayed, and participants were prompted to continue responding. Each task lasted for 180 s. To reduce the potential for creative engagement in the control task, participants were instructed to view each image and list the individual objects or people it contained, rather than describing the scene as a whole.

Participants completed the Epworth Sleepiness Scale prior to the experimental session, and only those who showed no signs of excessive daytime sleepiness were included in the main experiment. To minimize participant fatigue, all experimental sessions were conducted on a single day per participant. Each condition involved two 600 s vigilance tasks, separated by an intermediate task. To control for order effects, the sequence of the intermediate tasks — the Oogiri task and the control task — was counterbalanced across participants.

To standardize arousal levels at the beginning of each condition, participants first walked a predefined 230 m indoor route at their natural pace. This was followed by practice trials for both the

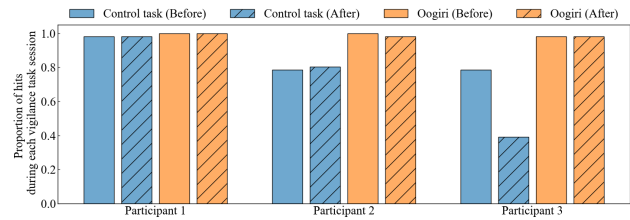


Figure 6: Proportion of hits in each vigilance task

vigilance and intermediate tasks. After the practice, participants rested for 120 s in a seated position. To minimize external influences on arousal, they were instructed not to engage in self-stimulation strategies such as inflicting pain or vocalizing. Subjective sleepiness was assessed before and after each vigilance and intermediate task using the Karolinska Sleepiness Scale. The study involved 3 male graduate students (mean age = 23.3 ± 0.94 years).

5.2 Experimental Result

Figure 6 reports the proportion of hits during each vigilance task session. Figure 6 presents a higher proportion of hits in the Oogiri task condition compared to the control task condition. This difference was evident even prior to the intermediate task, possibly due to the practice phase in which participants completed short versions of both tasks. Exposure to the Oogiri task during practice may have elevated or stabilized arousal levels. Although order effects, such as transient changes in sleepiness or accumulated fatigue, cannot be excluded, the data suggest that Oogiri improvisation may contribute to maintaining arousal and attentional performance over prolonged periods.

6 Conclusion

This exploratory study investigated whether an Oogiri task could mitigate drowsiness without altering environmental conditions or inducing discomfort.

The preliminary experiment confirmed that a vigilance task featuring target stimuli with a 10 % appearance rate induced drowsiness within 300 to 900 s. the proportion of hits and subjective assessments were found to be reliable indicators of arousal decline. In the main experiment, participants who engaged in the Oogiri task tended to maintain a higher proportion of hits during the subsequent vigilance task. These findings suggest that the task may support sustained attention by stabilizing arousal levels over time.

However, due to the small sample size, the findings should be interpreted with caution. Because all experiments were conducted in an environment with controlled auditory and visual input, future research should assess the generalizability of these findings to everyday settings, such as office environments, where environmental stimuli vary and task demands fluctuate. In addition, the development of a portable software application is needed to facilitate practical implementation in daily life.

Acknowledgments

This research is supported by Japan Tobacco Inc.

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